

### SECTION-D

**Note:** Long answer type questions. Attempt any one question out of two questions. (1x10=10)

- Q.19 Explain in detail the different types of cables used in electrical and communication systems, along with their applications.
- Q.20 Describe the importance of ESD safety in the workplace and the proper way to wear ESD clothing.

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**Level-5, 1st Sem/ DVOC (Medical Image Tech.)**  
**Subject : Tools, Equipment & Safety Measures-I**

Time : 2 Hrs.

M.M. : 50

### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (5x1=5)

- Q.1 Which of the following cables is used for high-speed data transmission?
- a) Ribbon cable                      b) Optical fiber cable
- c) Twin-lead cable                      d) Armored cable
- Q.2 What is the primary function of a coaxial cable?
- a) Power transmission
- b) Audio & video signal transmission
- c) Water resistance
- d) Electrical grounding
- Q.3 Which type of connector is commonly used in networking?
- a) rca                                      b) rj45
- c) hdmi                                      d) xir

- Q.4 Which cable consists of multiple insulated conductors?
- a) twin axial cable
  - b) multi-conductor cable
  - c) twin-lead cable
  - d) shielded twisted pair cable
- Q.5 Why is ESD clothing important?
- a) To prevent electrostatic discharge from damaging electronic components
  - b) To provide physical protection against, mechanical injuries
  - c) To improve the appearance of the worker
  - d) To increase conductivity of the body

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 What is the main difference between shielded and unshielded twisted pair cables?
- Q.7 Name two advantages of fiber optic cables over traditional copper cables.
- Q.8 What is the function of an RJ45 connector?

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- Q.9 Why is grounding important in electrical cables?
- Q.10 What precautions should be taken while wearing ESD clothing?

### SECTION-C

**Note:** Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

- Q.11 Explain the structure and application of coaxial cables.
- Q.12 What are the different types of electrical connectors and their uses?
- Q.13 Describe the purpose and benefits of armoured cables.
- Q.14 What are the common safety measures to be followed when handling electrical cables?
- Q.15 What is the purpose of ribbon cables and where are they used?
- Q.16 Explain the working of shielded twisted pair cables and their advantages.
- Q.17 What are the different materials used in ESD clothing and why?
- Q.18 How does a metallic sheathed cable differ from a non-metalic sheathed cable?

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